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Department of Computer Science

Assignment-I

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Topic : Bio-Sensor

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BIO SENSOR

Overview of Biosensor:

A biosensor is known as an analytical device, which incorporates biologically derived recognition molecules, such as, enzyme, antibody, cell, aptamer, or single-stranded DNA with a transducer to convert a biological signal into an electrical signal.

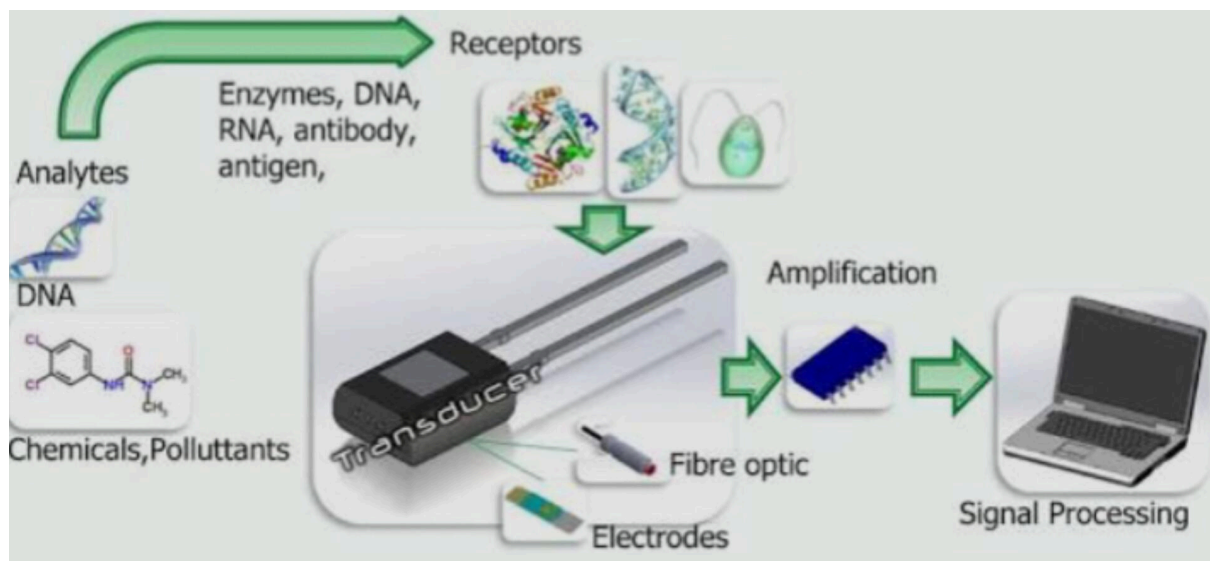


Fig:1

Basic Components of a Biosensor

1. **Bioreceptor** – A biological material (such as enzymes, antibodies, microorganisms, or DNA) that specifically recognizes the target analyte.
2. **Transducer** – Converts the biological interaction into a measurable signal (e.g., electrical, optical, or thermal).
3. **Signal Processor** – Amplifies, processes, and displays the signal in a readable form.

Types of Biosensors

Electrochemical Biosensors: Measure electrical signals produced by biological reactions (e.g., glucose sensors).

Optical Biosensors: Detect changes in light absorption, fluorescence, or refractive index.

Thermal Biosensors: Measure temperature changes resulting from biochemical reactions.

Piezoelectric Biosensors: Detect mass changes on a surface through variations in frequency or resonances

Advantages

- High sensitivity and specificity
- Real-time and rapid detection
- Compact and portable design
- Potential for miniaturization and integration with digital systems.

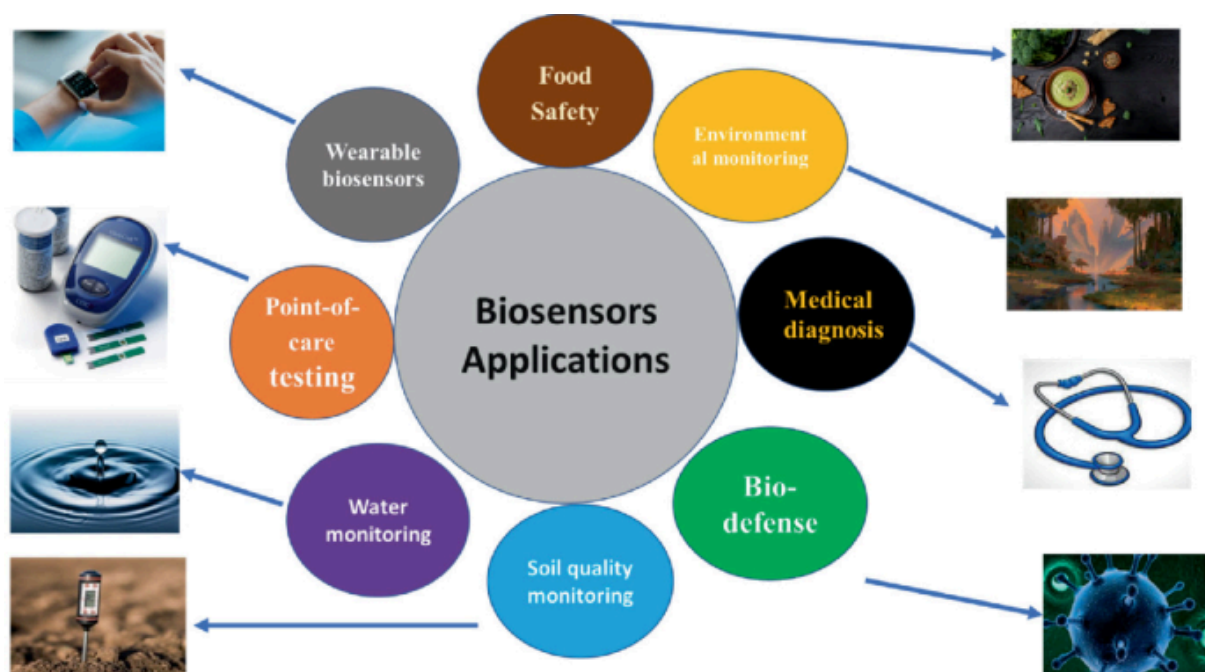


Fig:2

CASE STUDY 1: BIOSENSORS IN AUTOMATION

Introduction:

Biosensors enhance industrial automation by enabling real-time monitoring and control of processes through the detection of specific analytes.

Need for biosensor in automation:

Constraint In Traditional Automation	Biosensor Solution	Resulting Benefit
Traditional sensors cannot measure complex biological factors (e.g., how much nutrient a cell has consumed, or if a specific pathogen is present).	Biosensors are the only sensor capable of providing specific, real-time data on critical biochemical parameters (glucose, toxins, product yield).	Enables closed-loop control over dynamic bioprocesses, reducing reliance on manual testing.
Lab testing creates slow, expensive delays (off-line sampling).	Biosensors allow continuous, in-line monitoring.	Maximizes efficiency by preventing costly batch failures and ensuring process consistency and optimal product yield.

Selected Device: Lab-on-a-Chip / Microfluidic Biosensors.

Use: Automated lab testing, high-throughput analysis, portable automation systems.

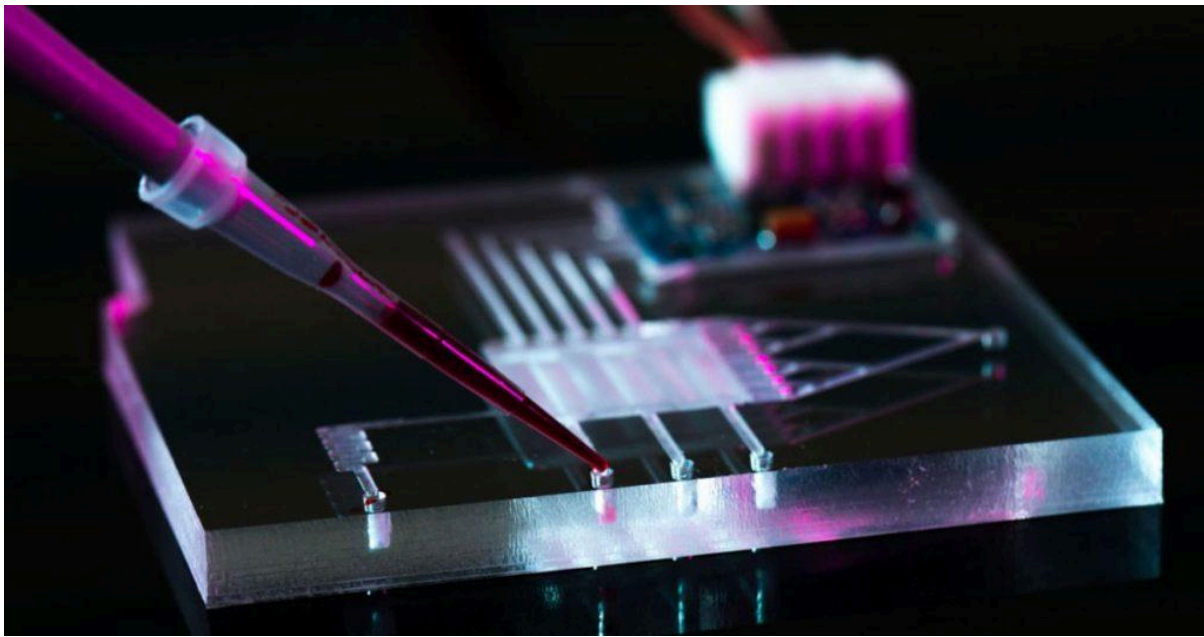


Fig:3. Lab-on-a-Chip / Microfluidic Biosensors

Device overview:

- Lab-on-a-chip (LOC) devices are miniaturized systems that integrate multiple laboratory functions, such as sample preparation and analysis, onto a single microchip. They use microfluidics to manipulate tiny fluid volumes in microchannels and offer advantages like faster analysis, lower costs, higher sensitivity, and portability compared to traditional labs. These devices are used in various applications, including diagnostics, environmental monitoring, and food safety.

Working Principle:

Lab-on-a-chip (LOC) devices work by using microfluidics to perform laboratory functions on a single chip, typically measuring millimeters in size. They manipulate and transport tiny volumes of fluids through microscopic channels, enabling processes like sample preparation, reaction, and detection to happen rapidly and automatically. This is achieved through the controlled movement of fluids via capillary action, pressure, or other forces, integrated with sensors to analyze the results.

Case Example: Automated Water Quality Monitoring Using LOC Biosensor

Application: Detecting heavy metals or pathogens in water using an LOC biosensor integrated with an automated sampler.

Automation Role:

- The system automatically collects samples at set intervals.
- The chip performs rapid testing (within minutes).
- Data is transferred to a computer or cloud-based system for decision-making.

Advantages:

- Requires microliters of sample & Fast analysis (seconds to minutes)
- Easy to integrate with IoT or robotic systems
- High accuracy and reproducibility & Portable and energy-efficient

Limitations:

- Fabrication cost.(for complex chips)
- Difficult to reuse.(single-use chips)
- Sensitive to environmental conditions. (temperature, humidity)



Fig:4.Automated Water Quality Monitoring Using LOC Biosensor

CASE STUDY 2 : Biosensors in Healthcare

Introduction:

Biosensors are devices that use biological materials to detect and measure chemical or biological substances. In healthcare, they are used for disease diagnosis, monitoring, and treatment. They provide quick, accurate, and real-time results, making them useful in glucose testing, infection detection, and wearable health monitors.

Importance of biosensor in Medical Fields:

Biosensors are vital in healthcare for early disease detection, continuous monitoring, and personalized medicine through their ability to rapidly and accurately measure biomarkers in bodily fluids.

Application of Biosensor in Healthcare:

- **Disease detection:** Biosensors can quickly detect pathogens, biomarkers, and genetic markers for diseases like cancer, infections (such as COVID-19), and cardiovascular conditions.
- **Chronic disease management:** They are crucial for managing conditions like diabetes by providing continuous glucose monitoring and for heart disease through heart rate and rhythm tracking.
- **Personalized medicine:** Biosensors facilitate real-time monitoring of a patient's response to medication, allowing doctors to optimize dosages and improve treatment efficacy, notes IntechOpen and MDPI.
- **Remote patient monitoring:** Wearable and implantable biosensors allow for continuous, remote tracking of patient data, improving access to care for those in remote areas or with mobility issues, explains the National Institutes of Health.
- **Drug delivery:** Biosensors can be integrated into closed-loop systems that automatically adjust drug delivery based on real-time physiological data, such as insulin pumps for diabetes management, say the National Institutes of Health.

Selected Device: COVID-19 rapid antigen test.



Fig:5.COVID-19 rapid antigen test.

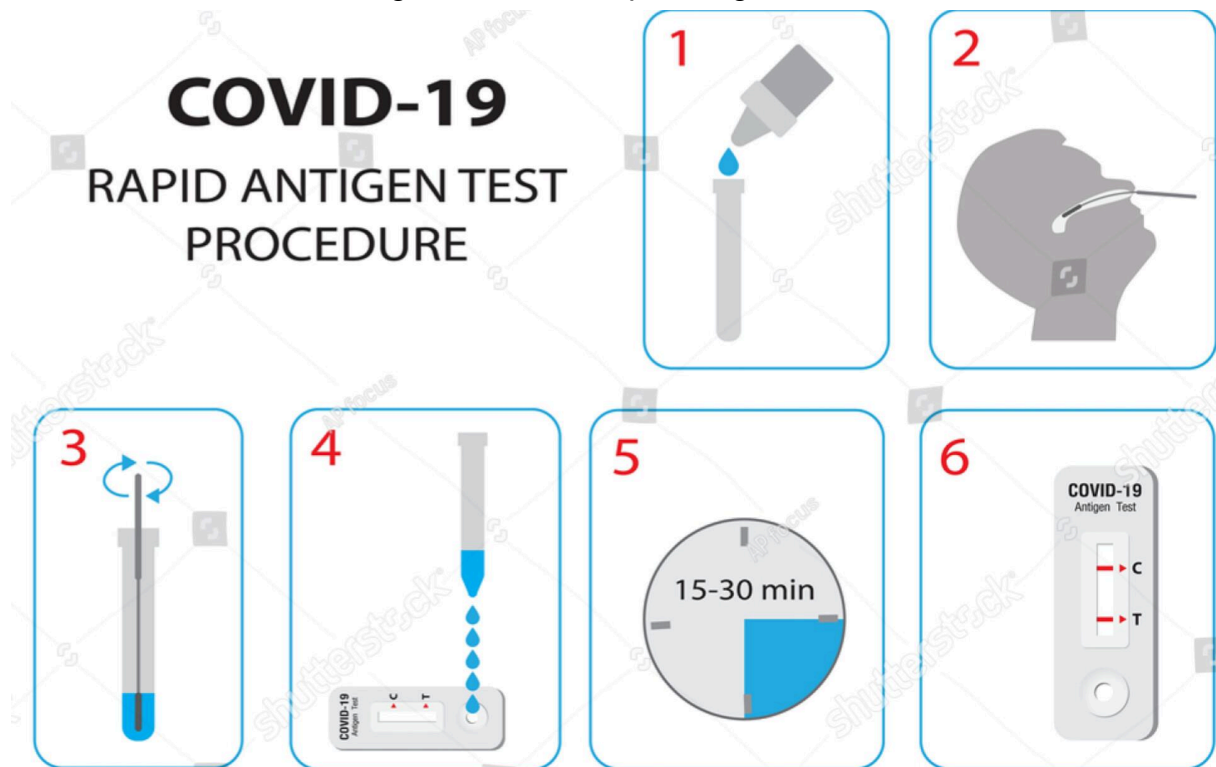


Fig:6

COVID-19 Rapid Antigen Test

The COVID-19 Rapid Antigen Test is a biosensor-based diagnostic test used to detect the presence of SARS-CoV-2 viral proteins (antigens) in a person's sample, usually collected from the nose or throat.

It works on the principle of an immunosensor, where antibodies on the test strip react with the viral antigens if present, producing a visible colored line (similar to a pregnancy test).

This test provides results within 15–30 minutes, making it ideal for quick screening and early detection of COVID-19, especially in hospitals, airports, and communities.

Example Device:

Abbott Panbio™ COVID-19 Ag Rapid Test Device – a widely used test kit that gives results in 15 minutes.

It detects the nucleocapsid protein antigen of the SARS-CoV-2 virus from nasal or nasopharyngeal swab samples.



Fig:7.Abbott Panbio™ COVID-19 Ag Rapid Test Device

Advantages of biosensor in healthcare:

- Fast and accurate results
- Portable and easy to use
- Real-time health monitoring
- Needs only a small sample
- Useful for early disease detection

Limitations:

- Expensive devices
- Short lifespan of sensors
- Accuracy issues sometimes
- Need regular calibration
- Data privacy concerns.

Impact:

Biosensors have greatly improved healthcare diagnosis and monitoring by providing faster, accurate, and real-time results. They help in early disease detection, better patient care, and support remote and home-based health monitoring. Overall, biosensors make healthcare more efficient, accessible, and personalized.

—Thank you—